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PROGRAMMING IN PYTHON

Project title: Health Habit Tracker

Submitted by:

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1. Introduction

Maintaining a healthy lifestyle has become increasingly difficult due to busy schedules, excessive screen usage, and lack of physical activity. Many individuals struggle to consistently track their daily habits, making it challenging to recognize patterns affecting their overall well-being.

Health Habit Tracker addresses this issue by providing a simple and user-friendly desktop application that allows users to record and analyze their daily health activities. The application offers an intuitive interface and automated summaries, enabling users to monitor their progress and stay motivated toward achieving healthier habits.

3. Objectives

The primary objectives of this project are:

1. To develop a desktop application for tracking daily health habits.
2. To provide an easy-to-use graphical interface using Tkinter.
3. To maintain persistent storage of records using JSON files.
4. To generate daily and weekly summaries using NumPy.
5. To provide health warnings based on predefined healthy ranges.
6. To demonstrate the use of Python programming concepts in a real-world application.

4. Scope of the Project

The application is intended for students, professionals, and individuals who wish to monitor their daily routines. It operates entirely offline and can be used on any computer with Python installed.

The project scope includes:

- Recording daily health metrics.
- Viewing, editing, and deleting records.
- Generating statistical summaries.
- Displaying health recommendations.
- Persisting data using local storage.

Future versions may include:

- User authentication.
- Cloud synchronization.
- Mobile application support.
- Graphical charts and analytics.
- Exporting reports to PDF or Excel.

5. Technologies Used

Technology	Purpose
Python	Core programming language
Tkinter	Graphical User Interface (GUI)
NumPy	Statistical calculations and summaries
JSON	Persistent data storage
OOP	Application architecture
File Handling	Reading and writing records

6. System Architecture

The application follows a modular architecture consisting of three major components:

Frontend

- Developed using Tkinter.
- Responsible for displaying forms, buttons, tables, and dialogs.

Backend

- Handles business logic.
- Performs validation and calculations.
- Generates summaries and health warnings.

Data Layer

- Stores records in `health.json`.
- Loads data automatically when the application starts.

Modules

1. `tracker.py`
 - Main application entry point.
 - Handles GUI operations.
2. `record.py`
 - Contains the `HealthRecord` class.
 - Implements Object-Oriented Programming principles.
3. `utils.py`
 - Provides helper functions.
 - Performs validation and summary generation.
4. `health.json`
 - Stores user records persistently.

7. Functional Requirements

The system shall provide the following functionalities:

- Add a new health record.
- View all records.
- Edit existing records.
- Delete records.
- Display daily summaries.
- Display weekly summaries.
- Show health warnings.
- Save records automatically.

8. Non-Functional Requirements

Requirement	Description
Performance	Application should respond quickly.
Reliability	Data should not be lost between sessions.
Usability	Interface should be simple and intuitive.
Maintainability	Code should be modular and easy to update.
Portability	Application should run on any platform supporting Python.

9. Features

Add Record

Users can enter:

- Date
- Sleep Hours
- Water Intake
- Exercise Minutes
- Screen Time
- Study Hours

The application validates the entered data before saving.

View Records

Displays all records in a table format, allowing users to review previously entered data.

Edit Record

Users can modify existing entries through a dedicated update window.

Delete Record

Records can be permanently removed after user confirmation.

Daily Summary

Shows the complete health information for a selected date.

Weekly Summary

Calculates:

- Average sleep hours
- Total water intake
- Average screen time
- Total exercise duration
- Average study hours

NumPy is used to perform these computations efficiently.

Health Warnings

Examples:

- Sleep less than 7 hours.
- Water intake below 8 glasses.
- Exercise less than 30 minutes.
- Screen time exceeds 8 hours.
- Study hours exceed recommended limits.

10. Use Case Diagram

Actors:

- User

Use Cases:

- Add Record
- View Records
- Edit Record
- Delete Record
- View Daily Summary
- View Weekly Summary
- Receive Health Warnings

11. Implementation Details

Object-Oriented Programming

The project uses a `HealthRecord` class containing:

- Date
- Sleep Hours
- Water Intake
- Exercise Minutes
- Screen Time
- Study Hours

File Handling

- Reads data from `health.json`
- Writes updates automatically
- Maintains persistence across application sessions

Exception Handling

Used to:

- Handle invalid input.
- Prevent application crashes.
- Ensure data consistency.

NumPy Integration

NumPy functions are used for:

- Mean calculations
- Sum operations
- Weekly statistical analysis

12. Testing

Test Case	Expected Result	Status
Add Record	Record saved successfully	Passed
Edit Record	Data updated correctly	Passed
Delete Record	Record removed	Passed
View Records	Displays all entries	Passed
Daily Summary	Correct summary shown	Passed
Weekly Summary	Accurate calculations	Passed
Invalid Input	Error message displayed	Passed

14. Challenges Faced

- Designing an intuitive GUI.
- Managing JSON file storage.
- Implementing input validation.
- Integrating NumPy calculations.
- Ensuring modular code organization.

15. Future Enhancements

Potential improvements include:

- Multi-user support.
- Database integration (SQLite/MySQL).
- Authentication system.
- Graphical analytics dashboard.
- Email reminders and notifications.
- Cloud backup support.
- Mobile application version.

16. Conclusion

Health Habit Tracker successfully demonstrates how Python can be used to build practical desktop applications. By combining Tkinter for the user interface, NumPy for data analysis, and JSON for persistent storage, the project provides an effective solution for tracking daily health habits.

The application encourages users to maintain healthier lifestyles by offering meaningful insights into their daily routines. Furthermore, the project serves as an excellent example of applying Object-Oriented Programming, file handling, exception handling, and GUI development concepts in a real-world scenario.